Model Assessment (Evaluation/Validation) and Model Selection Part 2

- Correct use of cross-validation methods for model selection and/or estimation of risk (assessment)
- Simple concepts with a rich formal notation, they can help to avoid ambiguities:
- Very useful guide. But it is not intended as a recipe book but to help you to rationalize, i.e. to understand in a systematic form issues of a rigorous approach.
- Then you have to be *reasonable (when you are aware*) (which effort deserve? Most of the *time* for running the experiments depends on the validation choices! We have to avoid both a too rough evaluation and a never ending evaluation process)
- In any case use them if you have uncertainties, it is better than to use *fantasy*!
- Note that we will discuss selection among different hyper-parameters configurations (represented by θ), but also for selecting among *different models* you have to properly apply "model selection" and "model assessment" approaches (think θ to represent also different models).

Memento: our exercise:

1) Estimation of error on TR and/or VL are NOT a good risk estimation.

2) Using the entire dataset for feature/model selection prejudices the estimation

Credits: by slides of S. Bengio, adapted and integrated by A. Micheli with the notation and figures for our course

Premise (refresh and slight changes in notations**)**

The data:

- Let $Z_1, Z_2, \dots, Z_l$ be a l-tuple random sample of an unknow distribution $p(z)$
- All the Z_i are independently and identically distributed (i.i.d.)
- Let D_r be a particular instance $= \{z_1, z_2, \dots, z_r\}$

Tasks:

- Classification: $Z = (X, Y) \in \mathbb{R}^n \times \{-1, +1\}$
- Regression: $Z = (X, Y) \in \mathbb{R}^n \times \mathbb{R}$

Loss: (to search for a good h in a function space H) (we called it "inner" loss function)

- Classification: $L(z, h) = L\big((x, y), h\big) = 0 \;\; if\ h(x) = y\, ;\ 1\ otherwise$
- Regression: $L(z, h) = L\big((x, y), h\big) = (h(x) - y)^2$

Aim: Minimize the expected risk (true error) on H, defined for a given h as

$$R(h) = E_Z[L(z, h)] = \int_Z L(z, h) p(z) dz$$

- Select $h^* = \arg\min_{h \in H} R\,(h)$
- Problems: $p(z)$ is unknow, and we don't have access to all $L(z, h)$

Empirical Risk : $R_{emp}(h, D_r) = \frac{1}{r}\sum_{p=1}^{r} L\big(z_p, h\big) = \frac{1}{r}\sum_{i=1}^{r} L(z_i, h)$
seen therein in a general sense as a *surrogate of R* on a finite set of data (D_r) with *r* data, within the purpose of training, model selection or model assessment as specified in the following algorithms, according to the used partition of data. *Don't* confuse it with the use in the SLT as training error, whereas, in SLT presentation, $R_{emp} = R_{emp}(h, D_{TR})$ according to the notation of this document.

Methodology

Identify the goal. It could be:

1. To give the best model you can obtain given a (*training*) data set?
2. To give the expected performance of a model obtained by the Empirical Risk Minimization given a (*training*) data set?
3. To give the best model and its expected performance that you can obtain given a (*training*) data set?

For (1): you need to do model selection
For (2): you need to do model assessment (i.e. to estimate the risk)
For (3): you have to do both

We will systematically see in the following the different methods

Note: the provided original *training* data set will be used *by us* for the 3 different purposes

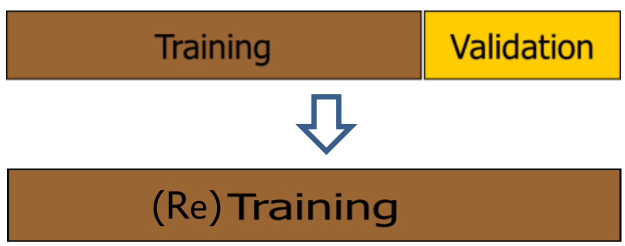
Model Selection: Hold-out Validation

- Select a hypothesis space with hyper-parameter θ
- Divide your dataset (training set) D_l in two parts
    - $D_{TR} = \{z_1, z_2, \dots, z_{tr}\},\ D_{VL} = \{z_{tr+1}, z_{tr+2}, \dots, z_{tr+vl}\},\ tr + vl = l$
- For each value θ_m of the hyper-parameter θ [grid search]
    - select $h^*_{\theta_m}(D_{TR}) = \arg\min_{h \in H_{\theta_m}} R_{emp}(h, D_{TR})$ [training]
    - estimate(#) $R\left(h^*_{\theta_m}\right)$ with $R_{emp}\left(h^*_{\theta_m}, D_{VL}\right) = \frac{1}{vl}\sum_{z_i \in D_{VL}} L\left(z_i,\ h^*_{\theta_m}(D_{TR})\right)$
- select $\theta^*_m = \arg\min_{\theta_m} R\left(h^*_{\theta_m}\right)$ [best on VL set]

Note: it does **not** return this value as estimation for test [(case 1) in the exercise]

- return $h^*(D_l) = \arg\min_{h \in H_{\theta^*_m}} R_{emp}(h, D_l)$ [retraining on l data]

(#)This estimation of R is only for model selection purpose (see note at page 2)

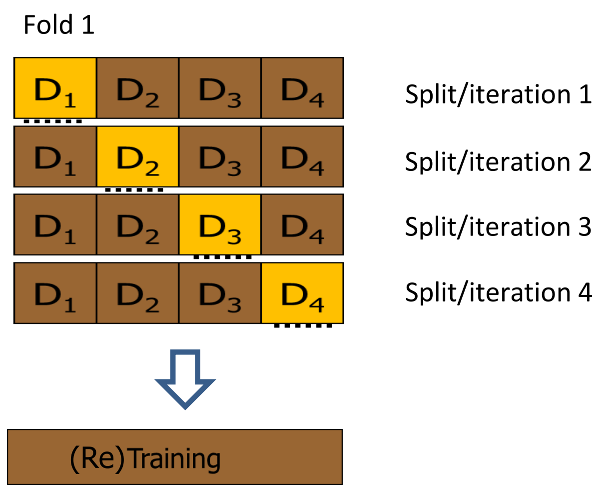


Model Selection: Cross Validation (K-fold CV)

- Select a hypothesis space with hyper-parameter θ
- Divide your dataset (training set) D_l into K distinct and equal parts $D_1, \ldots, D_k, \ldots, D_K$
- For each value θ_m of the hyper-parameter θ [grid search]
  - For each part D_k (and its counterpart $\bar{D}_k$) [folds cycle]
    - select $h^*_{\theta_m}(\bar{D}_k) = \arg\min_{h \in H_{\theta_m}} R_{emp}(h, \bar{D}_k)$ [training]
    - estimate $R\left(h^*_{\theta_m}(\bar{D}_k)\right)$ with $R_{emp}\left(h^*_{\theta_m}(\bar{D}_k), D_k\right) = \frac{1}{|D_k|}\sum_{z_i \in D_k} L\left(z_i, h^*_{\theta_m}(\bar{D}_k)\right)$
  - estimate(#) $R\left(h^*_{\theta_m}(D_l)\right)$ with $\frac{1}{K}\sum_k R\left(h^*_{\theta_m}(\bar{D}_k)\right)$
- select $\theta^*_m = \arg\min_{\theta_m} R\left(h^*_{\theta_m}(D_l)\right)$ [best on all θ_m values on the CVs]
- return $h^*(D_l) = \arg\min_{h \in H_{\theta^*_m}} R_{emp}(h, D_l)$ [retraining on l data]

Note the ORDER: Each iteration starts from scratch with training and estimation local to the split (else e.g. using a model trained on split 1 for split 2: you are using VL of split 2 that was used in training in split 1: the part D_2)

(#)This estimation of R is only for model selection purpose (see note at page 2)



Estimation of the Risk (model assessment): Hold-out

Note the ORDER:
the test set is separated NOW and not below: [(case 2) in the exercise]

- Divide your dataset (training set) D_l in two parts
  - $D_{TR} = \{z_1, z_1, \dots, z_{tr}\},\ D_{TS} = \{z_{tr+1}, z_{tr+2}, \dots, z_{tr+ts}\},\ \ tr + ts = l$

- select $h^*(D_{TR}) = \arg\min_{h \in H} R_{emp}\,(h, D_{TR})$ [training]

Note: it does not return here an estimation for test, but below [(case 1) in the exercise]

(this optimization process could include a model selection on the D_{TR} part)

- estimate $R\left(h^*(D_{TR})\right)$ with $R_{emp}(h^*(D_{TR}), D_{TS}) = \frac{1}{ts}\sum_{z_i \in D_{TS}} L\big(z_i,\ h^*(D_{TR})\big)$ [test error]

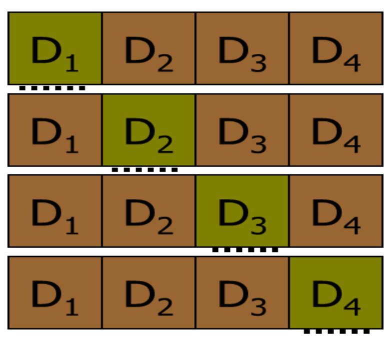
Estimation of the Risk (model assessment): Cross-validation (K-fold CV)

- Divide your dataset (training set) D_l into K distinct and equal parts $D_1, \ldots, D_k, \ldots, D_K$
  - For each part D_k [folds cycle]
    - let the counterpart $\overline{D}_k$ be the set of examples that are in D_l and not in D_k
    - select $h^*(\overline{D}_k) = \arg\min_{h \in H} R_{emp}(h, \overline{D}_k)$ [training]

      (this process could include a model selection)
    - estimate $R\left(h^*(\overline{D}_k)\right)$ with [test error in each iteration]
$$R_{emp}(h^*(\overline{D}_k), D_k) = \frac{1}{|D_k|} \sum_{z_i \in D_k} L\big(z_i, h^*(\overline{D}_k)\big)$$
  - estimate $R\left(h^*(D_l)\right)$ with $\frac{1}{K} \sum_k R\big(h^*(\overline{D}_k)\big)$ [the mean over the folds]

Notes: when $K = l$: *leave-one-out* CV

We do not violate the principia: the test set results are never used for training or model selection in any K-fold CV splitting/iteration



Merge: Model selection + Model assessment (estimation of the Risk)

- When you want both the best model and its expected risk.
- You then need to merge/combine the methods already presented.
- For instance:
  - hold out with training-validation-test: 3 separate data sets are necessary
  - cross-validation + test: cross-validate on training set, then test on separate set
  - external CV for test and internal hold out for the model selection
  - double-cross-validation: for each subset (of a K-fold CV), need to do a nested cross-validation with the $K - 1$ other subsets

- Other aspects:
  - take care to consider the different trials (different weigths *init* for a NN) when computing the mean values for TR, VL, TS

- More general methodological aspects:
  - compare your results with other methods
  - use statistical tests to verify significance
  - verify your model on more than one datasets

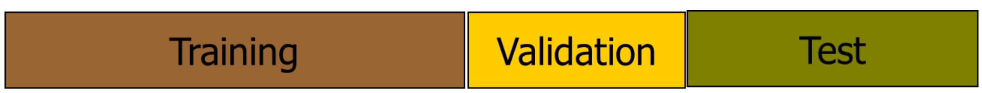
Merge: Training – Validation – Test (Hold Out)

- Select a hypothesis space with hyper-parameter θ
- Divide your dataset (training set) D_l into three parts D_{TR}, D_{VL}, D_{TS} , $tr + vl + ts = l$
- For each value θ_m of the hyper-parameter θ [grid search]
  - select $h^*_{\theta_m}(D_{TR}) = \arg \min_{h \in H_{\theta_m}} R_{emp}(h, D_{TR})$ [training]
  - let $R_{emp}\left(h^*_{\theta_m}(D_{TR}), D_{VL}\right) = \frac{1}{vl} \sum_{z_i \in D_{VL}} L\left(z_i,\ h^*_{\theta_m}(D_{TR})\right)$

- select $\theta^*_m = \arg \min_{\theta_m} R_{emp}\left(h^*_{\theta_m}(D_{TR}), D_{VL}\right)$ [best on VL set]
- select $h^*(D_{TR} \cup D_{VL}) = \arg \min_{h \in H_{\theta^*_m}} R_{emp}(h, D_{TR} \cup D_{VL})$ [retraining]

- estimate $R\left(h^*(D_{TR} \cup D_{VL})\right)$ with $\frac{1}{ts} \sum_{z_i \in D_{TS}} L\left(z_i,\ h^*(D_{TR} \cup D_{VL})\right)$

- Question: Can we now retrain the model again with all the D_l data? Does it use the test results?

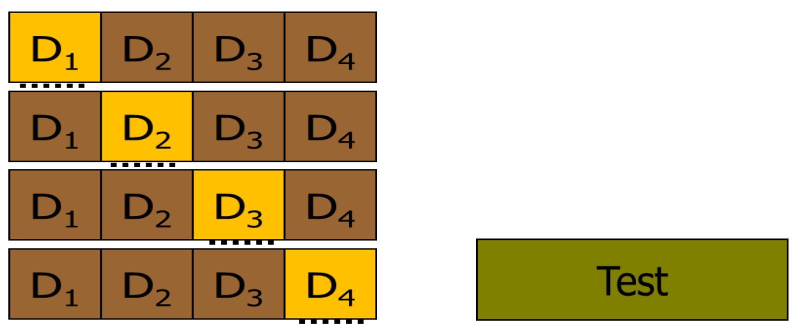


(retraining is not shown, it can have a VL set for the Early Stopping)

Merge: K-fold CV for model selection + Hold Out Test

- Select a hypothesis space with hyper-parameter θ
- Divide your dataset D_l in two parts D_{TR} and D_{TS}, $tr + ts = l$
- For each value θ_m of the hyper-parameter θ [grid search]
  - estimate[(#)] $R\left(h^*(D_{TR})\right)$ with D_{TR} using a K-fold CV
- select $\theta_m^* = \arg\min_{\theta_m} R\left(h_{\theta_m}^*(D_{TR})\right)$ [best from the K-fold CVs]
- select $h^*(D_{TR}) = \arg\min_{h \in H_{\theta_m^*}} R_{emp}(h, D_{TR})$ [retraining]
- estimate $R\left(h^*(D_{TR})\right)$ with $\frac{1}{ts}\sum_{z_i \in D_{TS}} L\left(z_i, h^*(D_{TR})\right)$

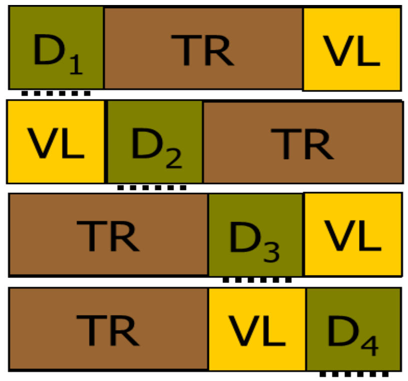
(#)This estimation of R is only for model selection purpose (see note at page 2), and this is the validation error over the K-fold CV on the "design set" called therein D_{TR}



(retraining is not shown, it can have a VL set for the Early Stopping)

Merge: K-fold CV for Test with internal Hold-out for model selection

The description by the meta algorithm can be an Exercise



(retraining is not shown)

The internal TR/VL Hold Out in each raw can be with arbitrary partitions and with shuffling (the figure is just illustrative)

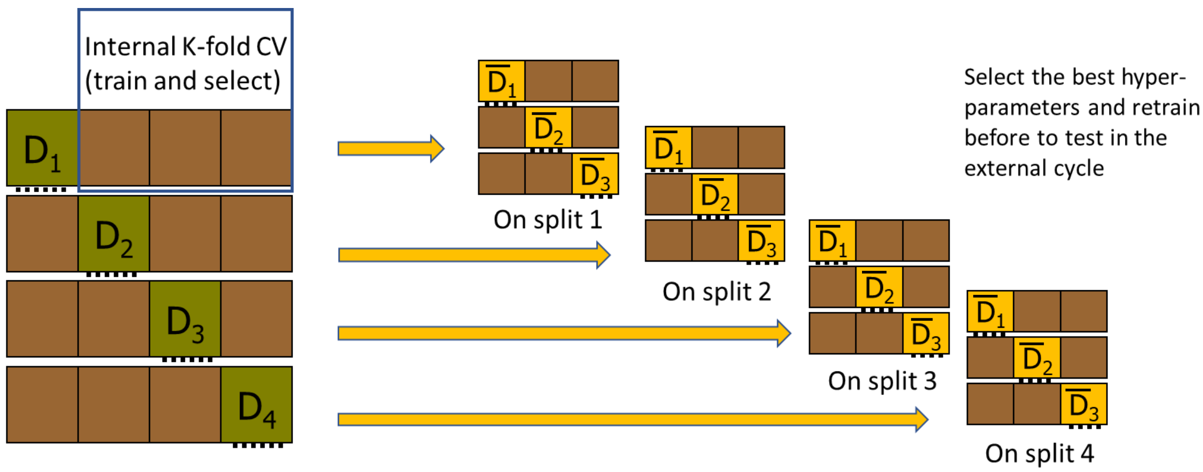
Note: this process **does not** provide a final model yet, it only gives an estimate of the risk (of the class of the considered models), but not a unique model (because you potentially have a different model for each raw). See the note in the following page.

Merge: Double K-fold CV (also known as Nested K-fold CV)

- Select a hypothesis space with hyper-parameter θ
- Divide your dataset (training set) D_l into K distinct and equal parts $D_1, \ldots, D_k, \ldots, D_K$
- For each part D_k (and its counterpart $\overline{D}_k$) [*external cycle*]
  - select $h^*(\overline{D}_k)$ by an internal K-fold CV on $\overline{D}_k$ (with K' that can be $\neq K$) [*inner cycle*]
  - estimate $R\left(h^*(\overline{D}_k)\right)$ with [test error for each external split]

$$R_{emp}(h^*(\overline{D}_k), D_k) = \frac{1}{|D_k|} \sum_{z_i \in D_k} L\left(z_i, h^*(\overline{D}_k)\right)$$

- estimate $R\left(h^*(D_l)\right)$ with $\frac{1}{K}\sum_k R\left(h^*(\overline{D}_k)\right)$ [the mean over the test folds]



(retraining in each internal cycle is not shown)

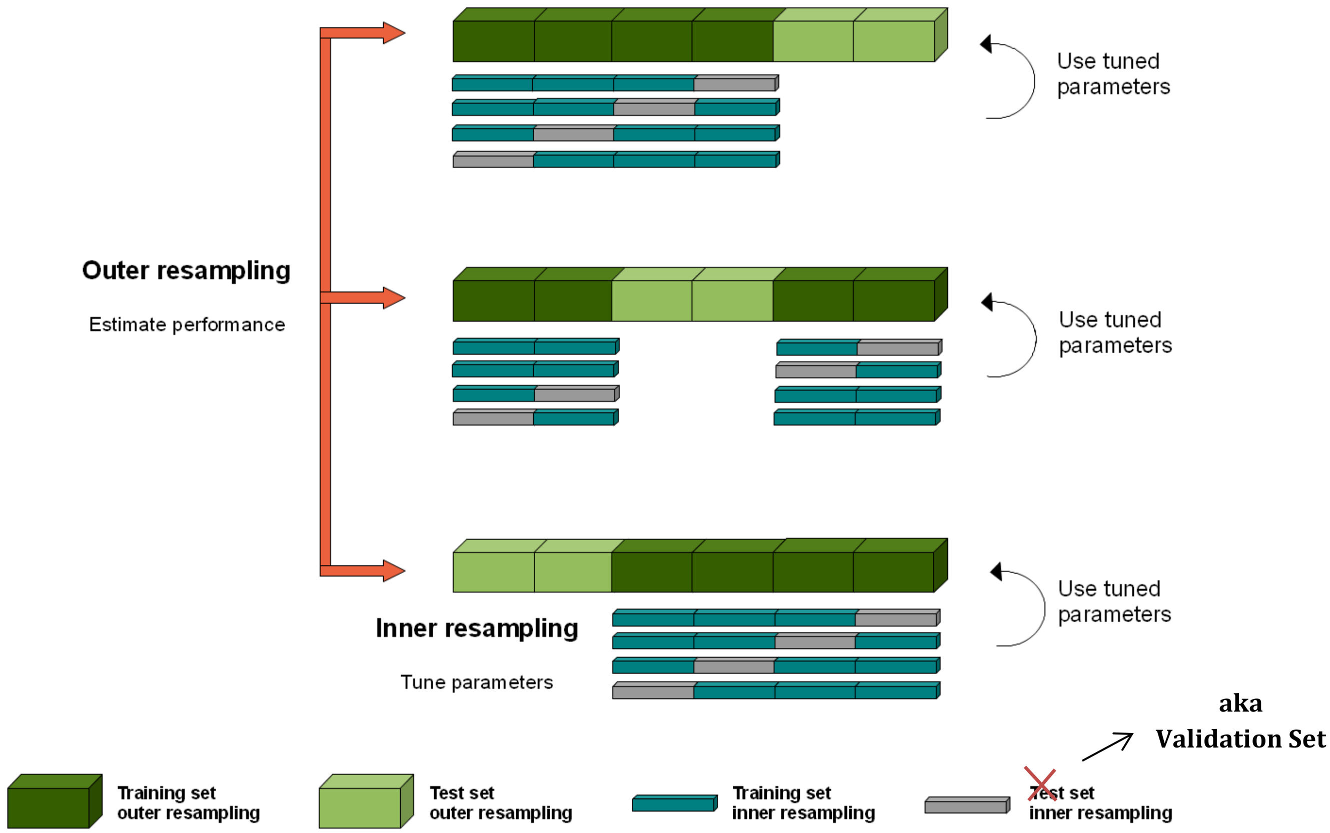
Note: this process **does not** provide a final model, it *only* gives an estimate of the risk, but not a unique model, because you potentially have a different hyperparameters for each external cycle step (external split).

If you need a final unique model, you can perform a **separate** model selection process (hold out or K-fold CV)! And a possible final retraining.
This approach does not violate the rules; the *test error has been already estimated* above (for the class of model/algorithm), with also the estimation of the variance (standard deviation - std) through the folds. We are <u>not</u> (and never) using the <u>test results</u> for any model selection, and the model will have an expected error within the estimated interval.

Moreover, note that for this reason it is not simply to say that the test happens *always* as a final step, in this case it is before the last model selection phase (even if it can appear a bit anti-intuitive).

Finally, note that the double CV makes *better exploitation of all the data* for training and evaluation (both for selection and assessment aims) even if at the cost of a *high computational effort*!

Double CV (a different view)



Nested/Double CV (a further different view, also with the inner Hold Out case)